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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester MCA (CBCS) Degree Examinations
May – June 2025

22MCA201 - DATA COMMUNICATION AND NETWORKS

Duration: 3 Hours

Max. Marks: 100

Note: Answer any **Five full** questions choosing **One full** question from **each Unit**

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the OSI model and its significance in computer networking. How does it help in standardizing communication?	10	L*2	1	1
	b) Explain the concept of Pulse Code Modulation (PCM) and differentiate between Asynchronous and Synchronous transmission.	10	L4	1	1
2.	a) Explain the different digital data encoding techniques. Discuss their advantages, limitations and apply it for data 1011011011.	10	L2	1	1
	b) Analyze the advantages and disadvantages of the TCP/IP model compared to the OSI model.	10	L2	1	1
Unit – II					
3.	a) Explain the concepts of flow control in data link layer communication. Compare and contrast Stop-and-Wait and Sliding Window flow control techniques.	10	L2	2	1
	b) Discuss how email is delivered over the internet. Compare the functionalities of POP3 and IMAP in final message delivery.	10	L6	2	1
4.	a) Explain the Domain Name System (DNS) and its role in the internet. Discuss the DNS name space, resource records, and the function of name servers in resolving domain names.	10	L6	2	1
	b) Discuss error control mechanisms in data link control. Explain how Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective-Reject ARQ work in ensuring reliable data transmission. Provide examples of their practical applications.	10	L6	2	1
Unit – III					
5.	a) Discuss the transport layer services provided to the upper layers of the OSI model. Explain the role of transport service primitives and how they facilitate communication between applications.	10	L6	3	1
	b) Describe the key elements of transport protocols, including addressing, connection establishment, and connection release. Explain how these elements ensure reliable data transfer in network communication.	10	L2	3	1
6.	a) Explain the structure and characteristics of the User Datagram Protocol (UDP). Discuss its common use cases. How does UDP support Remote Procedure Call (RPC) in network communication?	10	L6	3	1
	b) Discuss the Transmission Control Protocol (TCP) and its significance in reliable communication. Explain the TCP service model and its functionalities, including flow control, congestion control, and error handling.	10	L6	3	1

P.T.O.

Unit – IV

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| 7. a) | Discuss the concept of Store-and-Forward Packet Switching. How does it work, and what are its advantages and disadvantages in network communication? | 8 | L6 | 4 | 1 |
| b) | Compare and contrast the implementation of Connectionless Service and Connection-Oriented Service in the network layer. Explain how they function and provide real-world examples of each. | 12 | L2 | 4 | 1 |
| 8. a) | Explain Hierarchical routing algorithm with example. | 10 | L2 | 4 | 1 |
| b) | Differentiate between Virtual Circuit Subnets and Datagram Subnets. Discuss the advantages, disadvantages of each approach in networking. | 10 | L6 | 4 | 1 |

Unit – V

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| 9. a) | Explain the general principles of congestion control in networking. Discuss how congestion affects network performance and describe various congestion prevention policies used to mitigate it. | 10 | L6 | 5 | 1 |
| b) | Explain the concepts of Integrated Services (IntServ) and Differentiated Services (DiffServ) in network QoS management. Compare their working principles. | 10 | L5 | 5 | 1 |
| 10. a) | Compare and contrast congestion control mechanisms in Virtual-Circuit Subnets and Datagram Subnets. How do these approaches differ in handling congestion? | 10 | L2 | 5 | 1 |
| b) | Discuss the challenges in inter-networking due to differences between networks. Explain various techniques used to connect different networks, including Concatenated Virtual Circuits, Tunneling, and Fragmentation. | 10 | L6 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
